Shiva Shukla

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Summary

Software Engineer with 6+ years of experience building scalable, secure backend systems in Java Spring Boot, Spring WebFlux, Python FastAPI, SQL/NoSQL databases, and accessible front-end components with Lit and TypeScript. Delivered an MCP gateway with agents using Python FastMCP and LLM-based services on Azure, with expertise in microservices, REST APIs, OAuth2/JWT security, Redis cache, OpenTelemetry observability, Swagger, and automated testing. Experienced in CI/CD through GitHub Actions, GitLab, Jenkins, Helm Charts, Terraform, and Azure Kubernetes in Agile teams.

WORK EXPERIENCE

PTC Inc. | *Software Engineer Specialist (Full-Stack with Python, Java, TypeScript, Lit, Azure, SQL)* **Feb 2026 - Oct 2026**

- Built an Azure-based **Python FastMCP** gateway that exposes registered agent tools via tools/list and tools/call through **JSON-RPC 2.0** and **REST APIs**, with **JWT** and audience verification, request validation, timeouts, and standardized error handling. Added **Redis** and **in-memory caching** for an **Azure Storage**-backed registry to improve agent discovery and routing.
- Drove production readiness on Azure by integrating **OpenTelemetry** tracing, logging, and alerting into a **FastAPI** application for SLA/SLO compliance, with telemetry export (trace correlation, export-health monitoring, configurable **async-queue circuit breaker**) and an OTEL collector exporting to **Datadog** and **Prometheus**.
- Automated CI/CD pipelines using **GitHub Actions** and **GitLab** with environment-specific reusable components, integrating **HashiCorp Vault**, **Wiz**, **Black Duck**, and **Checkmarx**, and deploying services on **Azure Kubernetes** using **Helm charts** and pre- and post-deployment tests.
- Provisioned **Azure** infrastructure (AKS, Storage, Redis) as code with reusable **Terraform** modules in GitLab, cutting environment setup time by 20% and eliminating config drift across environments.
- Secured the Requirement Assistant (RA) AI tool plugin with **Spring Security** (OAuth2/JWT) and token rotation policies, reducing unauthorized requests by 15%.
- Improved test coverage for the RA tool from 30% to 80% with Playwright, JUnit, Pytest, and containerized integration tests with **GitHub Copilot, Claude**, and LLM-driven **prompt engineering**.
- Obfuscated 10+ core LLM algorithms using **Cython** to secure intellectual property for a private SaaS deployment, keeping latency within 3% of the Python baseline.

PTC Inc. | *Associate Software Engineer (Full-Stack with Python, Java, TypeScript, Lit, Azure, SQL)* **Aug 2023 - Jan 2026**

- Designed and developed localized, **WAI-ARIA-compliant TypeScript** and **Lit** web components for the RA AI tool in a software project management application, increasing accessibility by 50%.
- Developed the RA AI tool plugin using **Java, OOP, Spring Boot, RESTful APIs, Gradle, OAuth2/JWT, Tomcat, and Postgres** to mediate communication between the AI tool web component and backend **LLM** services, improving request processing throughput by 25%.
- Co-developed a **FastAPI** application using **LangChain** and **OpenAI LLMs** to evaluate/rewrite software requirements as per **INCOSE** standards, reducing token costs by 25% and improving F1 score by 22% through **prompt optimization**.
- Integrated an API in the RA tool Java plugin to send product and API metadata to **Snowflake** for faster failure triage.
- Consolidated BlackDuck, Checkmarx, and Invicti Jenkins pipelines for 5 product versions each, reducing redundancy by 80%.
- Optimized **stored procedures** in a **MySQL** database, improving the GET API performance by 30%.

GlobalLogic | *Software Engineer (Backend Engineer with Java, Spring Boot/WebFlux, NoSQL)* **Jul 2018 - Aug 2021**

- Developed **REST APIs** for a learning platform using **Java**, **Spring Boot**, **Reactive Streams**, **Kafka**, and the **Couchbase NoSQL** database; added unit tests with **JUnit**, **Mockito**, and **StepVerifier**, improving code quality by 35% and test coverage by 40%.
- Implemented **Kafka** with **Spring Cloud Streams API** and a retry mechanism to improve microservice reliability by 30%.
- Optimized **Couchbase N1QL** queries with composite indexes, improving lookup by 25% across 5 high-traffic REST endpoints.
- Authored consumer contracts using **Spring Cloud Contracts**, reducing early requirement gaps by 20%.

SKILLS

- **Languages**: Java, Python, JavaScript, TypeScript, SQL
- **Cloud**: Azure Kubernetes Service, Azure Storage, AWS – S3, EC2, RDS, DynamoDB, Lambda, CloudWatch
- **Frameworks**: Hibernate, Spring Boot, Spring Cloud, JUnit, Mockito, TestNG, ShellSpec, Lit, Playwright, FastAPI
- **Security**: OAuth2, JWT, Spring Security, HashiCorp Vault, Wiz, BlackDuck, Checkmarx, Invicti, SonarQube
- **Databases**: Couchbase, MySQL, PostgreSQL, Redis
- **AI**: Model Context Protocol (MCP), FastMCP, AI Agents, LangChain, LLM, Prompt Engineering, Claude, GitHub Copilot
- **Tools and Technologies**: Gitlab, Jenkins, Git, Docker, Kubernetes, Apache Tomcat, Kafka, Zookeeper, Maven, Gradle, Agile, Linux, Microservices, Helm, Terraform, OpenTelemetry, Prometheus, Datadog, Snowflake, Restful API, Swagger, GitHub Actions

EDUCATION

George Mason University **Aug 2021 - May 2023**
Master of Science, Computer Science - 3.93 *Fairfax, VA*

Harcourt Butler Technological Institute **Aug 2014 - Jun 2018**
Bachelor of Technology, Information Technology - 3.8 *Kanpur, India*